MVC (Model-View-Controller) Pattern
Shop-Ez E-Commerce Application

1. Model Layer (Data Layer)

• Handles database interaction and data-related logic.

• Defines schemas using Mongoose.

• Performs CRUD operations (Create, Read, Update, Delete).

• Interacts directly with MongoDB.

2. Controller Layer (Business Logic Layer)

• Acts as a bridge between Routes and Models.

• Processes incoming requests.

• Applies validation and business logic.

• Calls model functions and returns responses.

3. View Layer (Routing Layer)

• Defines API endpoints (GET, POST, PUT, DELETE).

• Handles HTTP requests.

• Invokes appropriate controller functions.

• Sends response back to the client.

Advantages of MVC Architecture

• Separation of Concerns.

• Improved Maintainability.

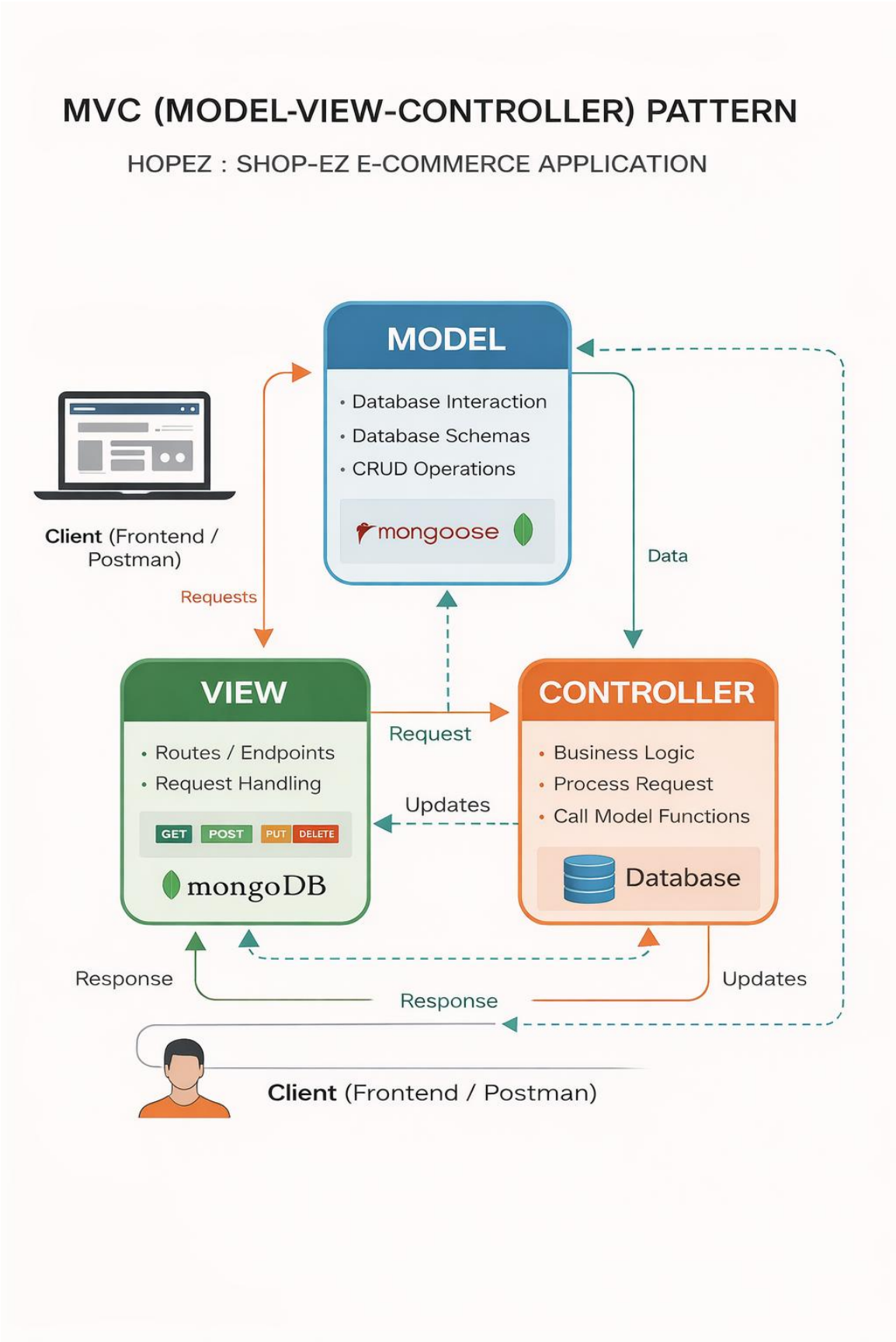
• Scalability.

• Code Reusability.

• Easy Testing.

• Collaboration Friendly Structure.

MVC Diagram:


MVC (MODEL-VIEW-CONTROLLER) PATTERN
HOPEZ : SHOP-EZ E-COMMERCE APPLICATION
MODEL
• Database Interaction
• Database Schemas
• CRUD Operations
mongoose
Client (Frontend / Postman)
Requests
Data
VIEW
• Routes / Endpoints
• Request Handling
GET
POST
PUT
DELETE
mongoDB
Request
Updates
CONTROLLER
• Business Logic
• Process Request
• Call Model Functions
Database
Response
Response
Updates
Client (Frontend / Postman)